PRE-FREEZE / DO NOT FABRICATE

PWR-2 rev.1 research schematic — 1S2P 18650 + BMS + MAX17048 + TPS61088/+12 V + LMR33630 IBB/−12 V + TPS61088/+5 V.
Charger/power-path (BQ24074) stays on CORE-2; PWR-2 converters are fed ONLY by VSYS (BQ24074 OUT). No MC34063, no charge pump.
Docs: README.md, power-tree-pwr2.md, interfaces.md, docs/schema/pwr2.md

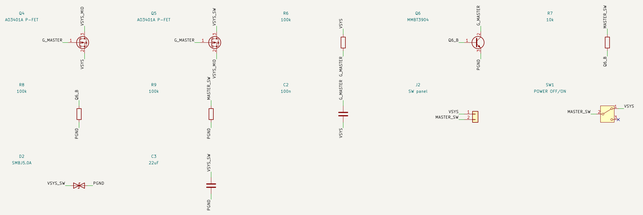
PACK 1S2P: 2x18650 parallel, J1 pack, F1 5 A, Q1 reverse-FET, D1 TVS, TH1 NTC



BMS 1S: BQ297xy + 2x AO3400A (common source), OVP/UVP/OCP/SCP; VBAT = P+ -> BQ24074 BAT



MASTER LOAD SWITCH: back-to-back AO3401A, MMBT3904 high-side driver, panel switch J2/SW1, TVS



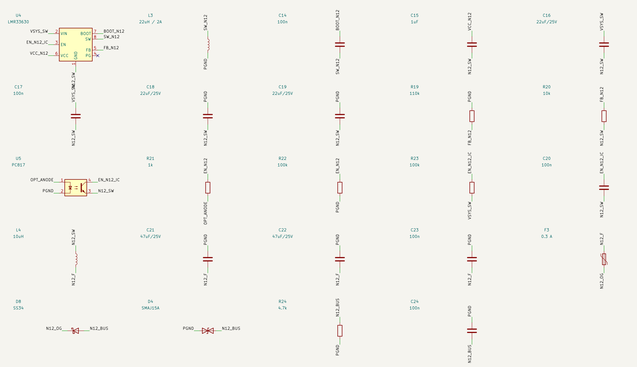
FUEL GAUGE MAX17048 (I2C 0x36) + Kelvin sense at the cell + analog backup BATT_MON



BUS +12 V: TPS61088 sync boost 12.15 V -> C-L-C pi-filter -> polyfuse 0.5 A -> TVS (budget 330 mA)



BUS −12 V: LMR33630 IBB (GND MC = −12 V, VIN = VSYS+12 V) + opto EN -> pi-filter -> polyfuse 0.3 A (budget 95 mA)



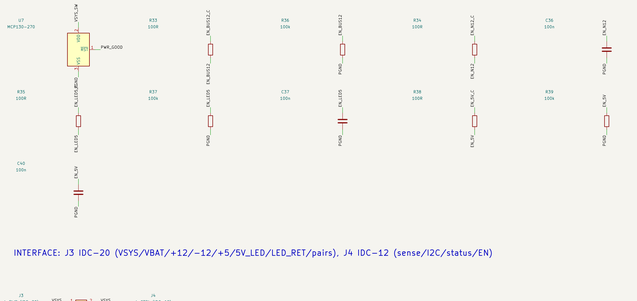
LOGIC/AUDIO +5 V: TPS61088 sync boost 5.10 V -> pi-filter -> polyfuse 2 A (budget 1.0 A)



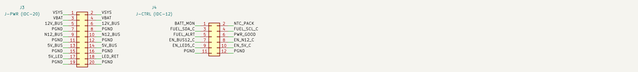
LED BRANCH: polyfuse 2.5 A + AO3400A low-side switch on the LED return (EN_LED5, OFF by default)



SUPERVISOR + EN INPUTS: MCP130-270 (PWR_GOOD open-drain), 100R/100n EN filters (3.3 V-safe)



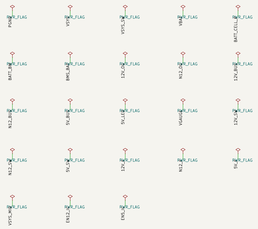
INTERFACE: J3 IDC-20 (VSYS/VBAT/+12/−12/+5/5V_LED/LED_RET/pairs), J4 IDC-12 (sense/I2C/status/EN)



TEST POINTS TP1..TP9



POWER FLAGS (net drivers for ERC)



Bukiton open hardware — CERN-OHL-S-2.0

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Title: PWR-2 PRE-FREEZE / DO NOT FABRICATE — battery + bus rails (1S2P)

Size: A0	Date:	Rev: 1.0
KiCad E.D.A. 10.0.x		Id: 1/1